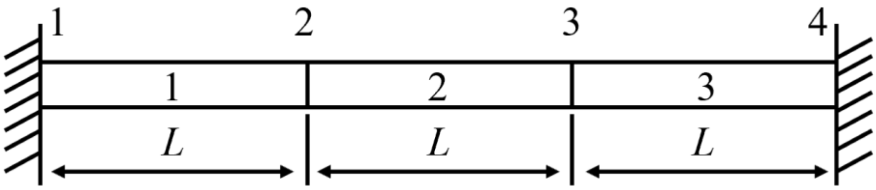
Variation Method & FEA Course
Homework 4

Submitted before June 22, 2026

1. The elastic rod is fixed at both ends. Suppose the section area is constant, the length is 3L and a body force f is uniformly applied. Divide the rod into 3 linear elements with the length of L and solve the element shape function matrix, element stiffness matrix and the global stiffness matrix.



2. Solve the following equations by the method of finite element. It is required to divide the interval into four elements by the nodes 0, 1/4, 1/2, 3/4 and 1.

$$-u''(x) + u(x) = -x$$
$$u(0) = u(1) = 0$$

Hints: The equivalent integral weak form of the equation is

$$Q[u(x)] = \frac{1}{2}\int_0^1 [u'^2(x) + u^2(x) + 2xu(x)]dx$$

3. Use the method of scraping line to directly satisfy the requirement of basis functions. All sides in the figure have equal lengths.